

Understanding Hallucinations in LLMs

Day 4 — Notebook 1 of 4 | Estimated Time: 25 minutes

What You'll Learn

- What hallucinations are and why they occur in language models
- The main categories of hallucinations (factual, reasoning, attribution)
- How to identify hallucinations through real examples
- Why hallucinations are an inherent property of how LLMs work

Setup

```
%pip install -U -q "google-genai>=1.0.0"
```

```
import sys, os
import time

import json
import re
from google import genai
from google.genai import types
from IPython.display import Markdown

try:
    from google.colab import auth
    auth.authenticate_user()
    print("✅ Authenticated via Colab")
except ImportError:
    print("✅ Using local credentials (gcloud auth)")

client = genai.Client(
    vertexai=True,
    project="chanakya-476010",
    location="us-central1",
)
MODEL_ID = "gemini-2.5-flash"
print("✅ Ready!")
```

```
✅ Authenticated via Colab
✅ Ready!
```

1. What Is a Hallucination?

A **hallucination** is when an LLM generates content that is:

- **Factually incorrect** — stating false information as fact
- **Fabricated** — inventing citations, names, dates, or events that don't exist
- **Inconsistent** — contradicting itself or the context provided
- **Ungrounded** — making claims not supported by any evidence

Why do LLMs hallucinate?

LLMs are trained to predict the **next most likely token** given a context. They don't "look up" facts — they generate text that *sounds plausible* based on patterns in training data.

Key causes:

1. **Training data gaps** — the model never saw accurate information on a topic
2. **Knowledge cutoff** — events after the training date are unknown
3. **Conflicting data** — contradictory information in training data

4. **Overgeneralization** — the model extrapolates incorrectly from patterns
5. **Instruction following pressure** — the model "tries to be helpful" even when it doesn't know the answer

2. Categories of Hallucinations

Category	Description	Example
Factual	Wrong facts about the world	Stating a historical date incorrectly
Attribution	Fake citations, quotes, or sources	Inventing a paper that doesn't exist
Entity	Wrong names, places, or identities	Misidentifying who wrote a book
Reasoning	Correct premise, wrong conclusion	Bad math or logical errors
Context	Ignoring or contradicting provided context	Answering a question the input didn't ask

3. Inducing a Factual Hallucination

Let's deliberately ask about a **fictional entity** that doesn't exist and observe the model's behavior.

```
# Ask about a completely fictional author to see if the model fabricates a response
response = client.models.generate_content(
    model=MODEL_ID,
    contents="""Tell me about the famous 19th century novelist Bartholomew Quincey-Finch
and his celebrated novel 'The Whispering Meridian' published in 1887."""
)
print(response.text)
```

Ah, Bartholomew Quincey-Finch, a name that evokes the very essence of late Victorian literary gravitas! While he and his celebra

Bartholomew Quincey-Finch (1845-1903)

Bartholomew Quincey-Finch was one of the more enigmatic and profoundly influential novelists of the late Victorian era, though h

Initially, Quincey-Finch pursued a brief and rather uninspired career in law, which he abandoned in his early thirties to dedica

Known for his reclusive nature and a rather austere public persona, Quincey-Finch rarely engaged in the literary circles of his

'The Whispering Meridian' (1887)

Published in 1887, 'The Whispering Meridian' is widely considered Bartholomew Quincey-Finch's magnum opus and a pivotal work tha

****Plot Summary:****

The novel unfolds primarily within the claustrophobic confines of Blackwood Manor, a decaying country estate on the bleak Englis

Alistair carries with him not only exotic artifacts and strange, unsettling tales, but also a profound air of psychological dist

****Themes and Significance:****

'The Whispering Meridian' is a rich tapestry of themes that resonated deeply with its contemporary audience and continue to prov

- **The Shadow of Empire:**** Published at the height of British imperial power, the novel subtly critiques the psychological to
- **Psychological Realism:**** Quincey-Finch's true genius lies in his unflinching exploration of the human psyche. The novel is
- **Science vs. Superstition/Primitivism:**** Alistair Finch's ethnographic studies and his detached, almost clinical observatio
- **The Fragility of Order:**** The decaying manor house and the encroaching moors serve as powerful metaphors for the breakdown
- **Feminist Undertones:**** Eleanor Vance is a complex and proto-feminist character, struggling against societal constraints an

****Reception and Legacy:****

Upon its publication, 'The Whispering Meridian' was met with a mixture of awe and discomfort. Critics praised Quincey-Finch's un

Over time, its reputation solidified as a groundbreaking work that bridged the gap between the didacticism of earlier Victorian

Bartholomew Quincey-Finch and 'The Whispering Meridian' stand as a testament to the Victorian era's capacity for profound self-r

Observe: Did the model invent details? Or did it correctly say it doesn't know?

Better-calibrated models will refuse or express uncertainty. Older/smaller models often fabricate plausible-sounding but entirely false details.

4. Citation Hallucinations (Fabricated Sources)

One of the most dangerous hallucinations is **inventing academic papers or sources**.

```
# Ask the model to provide academic citations
response = client.models.generate_content(
    model=MODEL_ID,
    contents="""Provide 5 academic papers with full citations (authors, title, journal, year, DOI)
    about the effect of music tempo on workplace productivity."""
)
print(response.text)
```

Here are 5 academic papers focusing on the effect of music tempo on workplace productivity or related cognitive task performance

1. **Cassidy, J. W., & Macdonald, R. A. R. (2007).** The effect of background music tempo on arousal and task performance in a co
DOI: 10.1177/0305735607077873
Summary: This study investigated how different tempos of background music (slow, medium, fast) affected physiological
2. **Shih, Y. N., Chien, H. Y., & Chiang, F. Y. (2012).** The effect of music tempo on task performance and arousal: an ERP study
DOI: 10.1037/a0028247
Summary: Using event-related potentials (ERPs) to measure brain activity, this research explored how different music
3. **Rappo, C. M., & Axtell, R. L. (2008).** The Effects of Music Tempo on Cognitive Task Performance and Heart Rate. **Perceptu**
DOI: 10.2466/pms.106.3.791-799
Summary: This study examined the impact of varying music tempos (slow, medium, fast) on cognitive task performance a
4. **Reiners, M. C., & Routh, D. K. (1996).** Music tempo and task performance. **Psychological Reports, 79**(3c), 1111-1114.
DOI: 10.2466/pr0.1996.79.3.1111
Summary: This paper investigated the direct relationship between music tempo and various measures of task performanc
5. **Kim, S., & Im, S. (2020).** The effects of music tempo and task type on creative performance. **Creativity Research Journal**
DOI: 10.1080/10400419.2020.1798361
Summary: This research explored how music tempo (slow vs. fast) interacted with different task types (creative vs. a

Warning: Citations generated by LLMs often look real but have **fake DOIs, wrong authors, or don't exist at all**. Always verify any citation from an LLM using Google Scholar or PubMed.

5. Reasoning Hallucinations (Confident but Wrong)

LLMs can be **confidently wrong** on arithmetic and logic problems.

```
# Classic hallucination trap: the model may get this wrong
response = client.models.generate_content(
    model=MODEL_ID,
    contents="How many times does the letter 'r' appear in the word 'strawberry'?",
)
print(response.text)
```

The letter 'r' appears **3** times in the word 'strawberry'.

```
# Let's verify the actual count programmatically
word = "strawberry"
count = word.count('r')
print(f"Actual count of 'r' in '{word}': {count}")
print(f"Letters: {list(word)}")
```

Actual count of 'r' in 'strawberry': 3
 Letters: ['s', 't', 'r', 'a', 'w', 'b', 'e', 'r', 'r', 'y']

6. Context Contradiction Hallucinations

LLMs can sometimes ignore or contradict the context they are given.

```
# Provide a context with specific facts, then ask about something NOT in it
context = """
Employee record:
- Name: Sarah Chen
- Department: Engineering
```

```

- Start date: March 5, 2021
- Manager: David Park
"""

response = client.models.generate_content(
    model=MODEL_ID,
    contents=f""Based on the following employee record, what is Sarah's salary?

{context}""",
)
print(response.text)

```

Based on the provided employee record, **Sarah's salary is not listed.**

The record contains her name, department, start date, and manager, but no information about her compensation.

Good behavior: The model should say the salary is not provided in the context.

Hallucination: The model invents a salary figure.

Modern models like Gemini 2.5 Flash are well-calibrated here, but always test your specific use case.

7. Measuring Confidence vs. Accuracy

A key insight: **LLM confidence does not correlate with accuracy**. The model uses the same fluent, confident tone whether it's right or wrong.

```

# Ask for obscure facts that are hard to verify – watch the tone
questions = [
    "What is the boiling point of gold in Kelvin?",
    "What was the exact population of Constantinople in 1450 CE?",
    "How many words did Shakespeare write in total across all his works?",
]

for q in questions:
    response = client.models.generate_content(
        model=MODEL_ID,
        contents=q,
    )
    print(f"Q: {q}")
    print(f"A: {response.text.strip()}")
    print("-" * 60)

```

Q: What is the boiling point of gold in Kelvin?
A: The boiling point of gold is approximately **3129 Kelvin** (K).

This is calculated by converting its boiling point from Celsius (2856 °C) to Kelvin:
 $2856\text{ °C} + 273.15 = 3129.15\text{ K}$

Q: What was the exact population of Constantinople in 1450 CE?

A: It's impossible to give an **exact** population figure for Constantinople in 1450 CE. Modern census data simply didn't exist

However, historians generally estimate the population to be in the range of **40,000 to 70,000 inhabitants**.

This was a dramatic reduction from its peak population in earlier centuries (which could have been anywhere from 400,000 to 1,000,000).

* **Warfare and sieges:** Most notably the devastating Fourth Crusade in 1204.
* **Economic collapse.**
* **Plagues:** Including the Black Death, which repeatedly ravaged Europe and the Byzantine Empire.

As a result, much of the city within its massive walls lay desolate and sparsely populated, with large areas converted back to farmland.

Q: How many words did Shakespeare write in total across all his works?

A: Estimating the exact number of words Shakespeare wrote is a bit tricky due to variations in editions, spelling conventions, and punctuation.

884,647 words

Some estimates might go slightly higher, closer to **900,000 words**, depending on the edition used and whether minor textual elements are included.

To put it in perspective:

* His plays make up the vast majority of this total.
* The King James Bible, for comparison, has around 783,137 words.

So, Shakespeare's complete works are a truly monumental body of text!

Exercise 1: Spot the Hallucination

Ask the model to describe a **real but obscure topic** you know well. Identify any factual errors.

```
# Exercise 1: Test hallucination on a topic you know
# TODO: Pick a topic you personally know well (your city, your profession, a hobby)
# Ask Gemini a specific factual question about it
# Evaluate whether the answer is accurate

my_topic = "Calendar" # TODO: Replace with your chosen topic
my_question = "Give me the day of the week for 28th of every month in the calendar of 2010 to 2012" # TODO: Write a specific fact

response = client.models.generate_content(
    model=MODEL_ID,
    contents=my_question,
)
print(response.text)
# print()
print("Is this accurate? What did the model get wrong?")
```

Here are the days of the week for the 28th of every month from 2010 to 2012:

```
**Year 2010**
* January 28, 2010: **Thursday**
* February 28, 2010: **Sunday**
* March 28, 2010: **Sunday**
* April 28, 2010: **Wednesday**
* May 28, 2010: **Friday**
* June 28, 2010: **Monday**
* July 28, 2010: **Wednesday**
* August 28, 2010: **Saturday**
* September 28, 2010: **Tuesday**
* October 28, 2010: **Thursday**
* November 28, 2010: **Sunday**
* December 28, 2010: **Tuesday**

**Year 2011**
* January 28, 2011: **Friday**
* February 28, 2011: **Monday**
* March 28, 2011: **Monday**
* April 28, 2011: **Thursday**
* May 28, 2011: **Saturday**
* June 28, 2011: **Tuesday**
* July 28, 2011: **Thursday**
* August 28, 2011: **Sunday**
* September 28, 2011: **Wednesday**
* October 28, 2011: **Friday**
* November 28, 2011: **Monday**
* December 28, 2011: **Wednesday**

**Year 2012 (Leap Year)**
* January 28, 2012: **Saturday**
* February 28, 2012: **Tuesday**
* March 28, 2012: **Wednesday** (February had 29 days)
* April 28, 2012: **Saturday**
* May 28, 2012: **Monday**
* June 28, 2012: **Thursday**
* July 28, 2012: **Saturday**
* August 28, 2012: **Tuesday**
* September 28, 2012: **Friday**
* October 28, 2012: **Sunday**
* November 28, 2012: **Wednesday**
* December 28, 2012: **Friday**

Is this accurate? What did the model get wrong?
```

Exercise 2: Prompt for Self-Uncertainty

Modify a prompt to ask the model to express its **confidence level** and flag anything it's unsure about.

```
# Exercise 2: Ask the model to rate its own confidence
# TODO: Add instructions for the model to:
# 1. Answer the question
# 2. Rate its confidence (High / Medium / Low) on each fact
# 3. Flag any details it is uncertain about

question = "What are the key differences between BERT and GPT-3 in terms of architecture and use cases?"
```

```
# TODO: Modify the prompt to elicit confidence ratings
prompt = f"""
Give me the answer to the provided question
{question}. Rate your confidence scale from High to Low on each fact. Also anything you are unsure about highlight it
...
"""

response = client.models.generate_content(model=MODEL_ID, contents=prompt)
print(response.text)

---

### **Use Cases**

1.  **Primary Goal & Interaction Paradigm:**
    *   **BERT:** Primarily designed for text understanding and analysis. Its common use case involves fine-tuning the
        *   Confidence: High
    *   **GPT-3:** Primarily designed for text generation. Its strength lies in few-shot or zero-shot learning via sopl
        *   Confidence: High

2.  **Specific Task Suitability:**
    *   **BERT:** Excels in tasks requiring deep comprehension of existing text.
        *   Text Classification: Sentiment analysis, spam detection, topic categorization.
        *   Named Entity Recognition (NER): Identifying entities like names, locations, organizations.
        *   Question Answering (Extractive): Finding the exact answer span within a given text.
        *   Semantic Search: Understanding the meaning of queries to retrieve relevant documents.
        *   Confidence: High
    *   **GPT-3: Excels in tasks requiring fluent and coherent text production or complex reasoning through text manipulation
        *   Content Generation: Writing articles, stories, code, marketing copy.
        *   Summarization (Abstractive): Creating new summaries that rephrase the original text.
        *   Translation: Translating text between languages.
        *   Chatbots & Conversational AI: Generating human-like responses in dialogues.
        *   Code Generation: Writing code snippets from natural language descriptions.
        *   Creative Writing: Generating poetry, scripts, song lyrics.
        *   Confidence: High

3.  **Adaptability without Fine-tuning:**
    *   BERT: Typically requires fine-tuning for new tasks to achieve high performance. While some prompting can be done
        *   Confidence: High
    *   GPT-3: Highly adaptable to new tasks without fine-tuning (zero-shot or few-shot learning). This is one of its mo
        *   Confidence: High

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### **Summary of Key Differences:



| Feature           | BERT                                    | GPT-3                                           | Confidence |
|-------------------|-----------------------------------------|-------------------------------------------------|------------|
| Architecture      | Encoder-only                            | Decoder-only                                    | High       |
| Directionality    | Bidirectional                           | Unidirectional (Left-to-Right / Autoregressive) | High       |
| Primary Objective | Text Understanding (MLM, NSP)           | Text Generation (Causal Language Modeling)      | High       |
| Parameters        | Hundreds of millions                    | Hundreds of billions (e.g., 175B)               | High       |
| Primary Output    | Contextual Embeddings                   | Generated Text                                  | High       |
| Key Use Cases     | Text Classification, NER, Extractive QA | Content Generation, Summarization, Chatbots     | High       |
| Adaptation        | Primarily <b>Fine-tuning</b> for tasks  | Primarily <b>Prompting</b> (zero/few-shot)      | High       |



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Uncertainties/Nuances (Highlighted):

*   Both models are built upon the Transformer architecture, so that's a shared foundational element, not a difference.
    *   Confidence: High (on the shared foundation)
*   While BERT is primarily used for understanding and GPT-3 for generation, there are some overlaps or areas where one can be
    *   Confidence: Medium (on the extent of overlap in practical application without significant adaptation)
*   The terms "smaller" for BERT and "massive" for GPT-3 are relative. BERT was considered large when it came out. The defining
    *   Confidence: High (on the relative size, but acknowledging it's relative)
```

Key Takeaways

Insight	Implication
LLMs predict tokens, not facts	Fluency ≠ accuracy
Confidence is not calibrated	Never trust an LLM citation without verification
Knowledge cutoffs are real	Ask about recent events with caution
Context gaps lead to invention	If something isn't in context, the model may fabricate it
Better prompts reduce (but don't eliminate) hallucinations	Grounding is the real solution — see Notebook 2

 Further Reading

Resource	Type	Link
Survey of Hallucination in LLMs	Paper	arxiv.org/abs/2311.05232
LLM Hallucinations Explained	Video	IBM Technology (YouTube)
Gemini Safety & Reliability	Docs	ai.google.dev/gemini-api/docs

Next up: [02_Grounding_Techniques.ipynb](#)